

Revised Problem Statement

Multi-Task Analysis of Telugu-English Code-Mixed News Comments

1. Problem Statement

The rapid proliferation of Telugu-language news content on platforms such as YouTube has generated large volumes of user-generated comments that exhibit a distinctive linguistic phenomenon: Telugu-English code-mixing, colloquially referred to as Tenglish. In these comments, users fluidly alternate between Telugu script, Romanized Telugu, and English within a single utterance — for example: "CM garu cheppindi correct kadhu, complete fake agenda" or "bro idhi chala dangerous situation, meeru artham cheskovadam ledi."

This code-mixed register poses significant challenges for standard NLP pipelines, as existing models are trained on either monolingual Telugu or clean English text and fail to generalize to the mixed-script, informal, and often morphologically irregular nature of Tenglish social media text.

Despite the growing volume of such content and its societal relevance — particularly in the context of political discourse, misinformation reactions, and online toxicity — Telugu-English code-mixed text remains severely underrepresented in NLP research. Existing benchmarks (DravidianLangTech 2021–2024) have addressed offensive language detection in Tamil-English and Kannada-English to a greater extent than Telugu-English, highlighting the need for dedicated resources and benchmark studies for Telugu-English code-mixed news comments.

2. Research Objectives

This work addresses the following research questions, motivated by four specific gaps in existing literature:

1. To the best of our knowledge, no publicly available Telugu-English code-mixed YouTube news comment dataset jointly annotated for sentiment, toxicity, and stance currently exists.
2. Prior Telugu and code-mixed NLP studies primarily investigate sentiment, toxicity, and stance as independent tasks, with limited exploration of joint multi-task learning frameworks.

- 3. No systematic comparison of mBERT, XLM-R, IndicBERT, and MuRIL exists on Tenglish comment data across these three tasks.
- 4. Explainability analysis (SHAP/attention) on mixed-script Tenglish text remains unexplored.

RQ1. How effectively can multilingual transformer models (mBERT, XLM-R, IndicBERT, MuRIL) perform sentiment detection, toxicity detection, and stance detection on Telugu-English code-mixed news comments?

RQ2. To what extent does a multi-task learning framework — where all three tasks share a common transformer encoder — outperform single-task fine-tuned baselines?

RQ3. Which Telugu and English tokens most influence model predictions, as revealed by SHAP and attention-based explainability analysis?

RQ4. What is the impact of transliteration-aware normalization on downstream classification performance across the three tasks?

3. Proposed Tasks and Label Definitions

Task 1 — Sentiment Detection

Captures the overall emotional polarity expressed by the commenter.

Label	Definition	Example (Tenglish)
Positive	Comment expresses approval, praise, satisfaction, or optimism	"Super decision garu, chala bagundi"
Negative	Comment expresses disapproval, criticism, frustration, or pessimism	"Worst government, meeru em chestunnaru"
Neutral	Comment is factual, informational, or does not carry clear emotional tone	"Meeting rapu 3 PM ki undi ani chepparu"

Task 2 — Toxicity / Offensive Language Detection

Captures whether the comment contains harmful, abusive, or offensive language.

Label	Definition	Example (Tenglish)
Toxic	Contains explicit abuse, hate speech, threats, targeted harassment, or slurs	"Vadu oka [slur], vadiki em telidu"
Not Toxic	Does not contain offensive language, regardless of sentiment polarity	"Idi correct kaadu, better option choose cheyyali"
Ambiguous	Contains borderline language (mild profanity, sarcasm) where intent is unclear	"Haha typical politics ra"

Note: A comment can be Negative in sentiment without being Toxic. These are independent labels. A criticism is not automatically toxic.

Task 3 — Stance Detection

Captures the commenter's stance toward the news topic or event reported in the video.

Label	Definition	Example (Tenglish)
Support	Comment agrees with, defends, or endorses the news event, person, or policy discussed	"CM garu correct ga chepparu, anni facts unnai"
Oppose	Comment disagrees with, criticizes, or challenges the news event, person, or policy	"Idhi complete wrong decision, janalu suffer avutaru"
Neutral	Comment neither supports nor opposes; observational, informational, or topic-irrelevant	"Naaku emi artham kaaledu, ela untundo chudali"

Note on stance annotation: Stance is determined relative to the news topic inferred from the video title. Annotators will be provided the video title as context for each comment. A comment expressing general negativity without targeting the specific news event should be labeled Neutral on stance.

4. Annotation Protocol (Proposed)

- **Unit of annotation:** Individual comment (top-level + replies)
- **Context provided to annotator:** Video title + comment text

- **Labels per comment:** Three independent labels (Sentiment, Toxicity, Stance)
- **Ambiguous cases:** Use the Ambiguous / Neutral label; do not guess
- **Annotator:** Primary — Aladurthi Sai Meghanath; Secondary — TBD (for Cohen's κ on a 200-comment subset)
- **Agreement target:** Cohen's $\kappa \geq 0.70$ on the double-annotated subset
- **Tool:** Browser-based annotation tool (custom built)

A browser-based annotation tool has been developed to support the revised multi-task schema and provide video-title context during stance annotation.

5. Dataset Expansion Plan

Per Dr. Thushara's recommendation, the corpus will be expanded across the following news categories to improve diversity:

Category	Target channels / search terms
Politics	CM press meets, Assembly sessions
Social issues	Farmer protests, welfare schemes
Crime	High-profile cases, police encounters
Entertainment	Film releases, celebrity news
Sports	Cricket, Kabaddi, local sports

Current corpus: 2,316 comments **Expanded target:** 8,000–10,000 comments

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